

SEAL: Statistically-Bounded Erasure Audits for Machine Unlearning under Mutual Distrust

Quang-Vinh Dang

Abstract—A data owner who submits an erasure request to a machine learning system operator has no direct way to verify that the request was honored, and the operator has no incentive to volunteer proof of non-compliance. Prior work shows that any audit accurate enough to catch a dishonest operator necessarily leaks information about the retained population, but stops short of a mechanism a mutually distrusting pair can actually run. This paper presents SEAL, a two-channel erasure-audit certificate that separates a zero-privacy-cost check on the data owner's own forgotten records from a differential-privacy-bounded check on synthetic boundary probes, calibrated against replicated honest re-executions rather than assumed thresholds, and its counterpart SEAL-W for large language model unlearning, which retrieves a randomly chosen member of a paraphrased query cluster through real Paillier-based private information retrieval and cross-checks response diversity against a phantom-topic baseline. We give five theorems establishing exactly what each channel provably defeats – private information retrieval obliviousness under the decisional composite residuosity assumption, a hypergeometric bound on a resource-limited adversary's evasion probability, an exact finite-sample calibrated threshold, a zero-diversity guarantee against uniform response suppression, and a Bonferroni-corrected composition bound whose absence we found and fixed in our own implementation while writing this proof. We reproduce the verification methodology of WaterDrum, the closest prior watermark-based unlearning audit, as an explicit baseline, and show empirically that it is defeated by a server that detects and filters the audit query itself – the exact gap WaterDrum's own authors report as unsolved – while SEAL-W's diversity channel catches it. Experiments on federated logistic regression over UCI German Credit, MNIST, and a 150,000-row Forest Covertype subsample, and on a real fine-tuned language model audited over the real TOFU fictitious-author benchmark, report both power and honestly-measured failure modes at small calibration-sample sizes, rather than tuning them away.

Index Terms—Machine unlearning, differential privacy, private information retrieval, text watermarking, auditing, federated learning, right to be forgotten.

1 INTRODUCTION

Data-protection regulation gives individuals a right to have their data erased from systems trained on it [1]. When the system is a machine-learned model, “erasure” is a claim about a training procedure the data owner cannot observe: the model operator asserts that a retraining or fine-tuning step removed the record's influence, and the data owner must decide whether to believe it. This is not a benign estimation problem. The operator may find honest

unlearning expensive and have every incentive to claim compliance while changing nothing (*lazy noncompliance*), or to narrowly patch only the behavior an audit is expected to probe while leaving the record's broader influence untouched (*targeted spoofing*).

Two recent results sharpen why this is hard rather than merely inconvenient. Tang, Joshi, and Kundu give an information-theoretic argument that, for convex models, any audit relying only on the model's behavior (query-response pairs) cannot separate an honestly-unlearned model from an insufficiently-unlearned one without revealing membership information about the retained set [2] – there is an unavoidable trade-off between how powerful a behavioral audit is and how much it leaks about records the data owner was never entitled to see. Nguyen et al.'s veriFUL framework independently surveys the federated-unlearning-verification landscape and concludes that no proposed scheme yet gives a federation of mutually distrusting clients and servers a concrete, run-once verification protocol with a stated privacy cost [3]. Both papers characterize the problem; neither gives an auditor a mechanism to run.

1.1 Contributions

This paper's contribution is a concrete audit protocol – not a new impossibility result, and not a claim to have invented differential privacy, private information retrieval (PIR), or text watermarking. It is the specific combination of known primitives into a certificate that (i) makes the audit power/privacy-leakage trade-off in [2] a dial the data owner turns rather than an unavoidable fact they simply accept, and (ii) instantiates that dial concretely for two settings veriFUL leaves open: federated classical-ML unlearning, and large-language-model unlearning verified via watermarking. Concretely:

- 1) A two-channel certificate (SEAL) that separates a *zero-marginal-privacy-cost* channel – the data owner re-evaluates the model on data it already possesses in full, which costs no retained-set privacy at all – from a *tunable-privacy-cost* channel using Gaussian-mechanism-noised synthetic boundary probes, whose spend is set by a stated (ϵ, δ) budget (Section 3.2).
- 2) A calibration procedure that builds each channel's null distribution from literal re-executions of the

honest mechanism, matched to the real run’s base seed and per-client ordering, rather than an assumed parametric null – a detail we show empirically is not optional (Section 3.2).

- 3) SEAL-W, a watermark-based certificate for LLM unlearning that plants several independently-paraphrased queries about the same forgotten entity in a committed batch, retrieves one at random via real Paillier-based PIR, and adds an independent response-diversity channel targeting exactly the adversary class – uniform, topic-level output filtering – that a single-query watermark check structurally cannot catch (Section 3.3).
- 4) Five theorems with complete proofs (Section 4.1; full derivations in the supplemental material) establishing precisely, and only, what each channel is proven to defeat – including a composition bound whose violation we found in our own first implementation by writing the proof down, not by an experiment failing visibly, and subsequently fixed.
- 5) An explicit reproduction of WaterDrum’s verification methodology [4] as a first-class experimental baseline, and a demonstration that it is defeated by exactly the server behavior its own authors report as an open problem, while SEAL-W’s diversity channel is not (Section 4.2).
- 6) An empirical evaluation (Section 5) on three classical-ML datasets spanning three orders of magnitude in per-client data volume, and on a real fine-tuned language model audited over the real TOFU fictitious-author benchmark [5], reporting failure modes – including a benchmark-scale finding that audit power *degrades* as per-client data grows – as honestly as the successes.

Every individual primitive reused here – the Gaussian mechanism and zero-concentrated differential privacy (zCDP) accounting [6], Paillier-based PIR [7], the Kirchenbauer et al. green-list watermark [8], and the general oblivious-audit-via-PIR pattern of RESPIR [9] – is standard and cited rather than re-derived, per the guidance that theoretical derivations already published elsewhere belong in the reference list, not repeated. What is new is their specific composition into a certificate that answers a stated open problem in the unlearning-verification literature, and the formal accounting of exactly which adversary class each piece of that composition defeats.

2 RELATED WORK

2.1 Unlearning Verification under Mutual Distrust

Tang, Joshi, and Kundu’s geometric transfer theorem [2] is, to our knowledge, the sharpest existing formal statement of why this problem is hard: for convex models, it derives a per-point lower bound relating a behavioral audit’s accuracy to the leakage it must incur on a named retained point, and shows this bound is essentially tight for the class of audits it covers. It is a *characterization* result – it says what any sufficiently powerful behavioral audit must cost, not how to build one that spends that cost deliberately, chooses where to spend it, or composes multiple audit channels with

different leakage profiles. Nguyen et al.’s veriFUL survey [3] independently reaches the same practical conclusion from the federated-learning side: after cataloguing verification entities, goals, and metrics proposed across the federated-unlearning literature, it concludes that none of the surveyed schemes gives a federation of mutually distrusting clients and servers a concrete, run-once protocol with a stated, controllable privacy cost. Neither paper is in tension with SEAL’s design; SEAL is best read as an attempt at the mechanism both papers imply is missing, evaluated against the specific adversary classes (lazy noncompliance, targeted spoofing) that motivate their impossibility and survey arguments respectively.

González et al.’s sequential auditing framework [10] solves an adjacent but structurally different problem: given repeated, i.i.d. access to a mechanism under two auditor-chosen adjacent datasets, it gives an anytime-valid test of whether the mechanism satisfies a *claimed* (ϵ, δ) -DP guarantee, with dramatically fewer samples than earlier batch tests. This does not transfer directly to the erasure-audit setting: the data owner here does not control a repeatable mechanism it can re-run on chosen adjacent inputs (it gets one trained model), and there is no claimed (ϵ, δ) for the *training* process to audit against – the quantity SEAL bounds is the leakage of its own *audit protocol*, not a property of the server’s unknown, unaudited training pipeline. Table 1 summarizes how these three lines of work, plus Godinot et al.’s RESPIR (below), relate to SEAL/SEAL-W along the dimensions that matter for the mutual-distrust setting.

2.2 Watermark-Based Data Provenance

Kirchenbauer et al. introduce a keyed, context-seeded green/red vocabulary partition for LLM generation, detected via a z -test on the green-token fraction [8]; Lau et al.’s Waterfall generalizes this to a vocabulary-permutation-and-perturbation scheme applied via an LLM paraphraser, targeting robustness and scalability for large-scale text provenance rather than unlearning specifically [11]. Lu et al.’s WaterDrum applies this family of watermarks specifically to unlearning verification, and is the closest prior work to SEAL-W [4]: its data-centric metric watermarks a data owner’s contribution and checks for the watermark’s continued presence after a claimed unlearning step. Its own appendix explicitly states that a server which detects and filters the watermark check itself is not handled by their construction – Section 4.2 reproduces this gap directly, as an experimental comparison against our own implementation of their verification methodology, rather than only citing the limitation as stated.

2.3 Oblivious Auditing via Private Information Retrieval

Godinot et al.’s RESPIR uses lattice-based (LWE) PIR to let an auditor commit to a batch of candidate queries and retrieve a checked subset obliviously, with planted canaries as tamper sentinels, for fairness audits of a model provider that may adapt its answers once it can distinguish an audit query from a genuine one [9]. SEAL-W adopts this committed-batch-with-canaries pattern – the auditor commits the server to an answer for every candidate before revealing which will be checked – but substitutes Paillier-based, single-server

TABLE 1
Positioning SEAL/SEAL-W against the closest prior work.

Work	Setting	Runnable protocol?	Stated privacy budget?	Mutual distrust?	Crypto. oblivious?
Tang et al. [2]	Behavioral unlearning audit	No (impossibility bound)	Characterized, not chosen	Yes	N/A
Nguyen et al. [3]	Federated unlearning (survey)	No (identifies the gap)	Not addressed	Yes	N/A
González et al. [10]	Auditing a claimed DP guarantee	Yes, for repeatable mechanisms	Type-I error, not audit leakage	No (chosen datasets)	N/A
Godinot et al. [9]	Fairness audits, deceptive model	Yes	Not privacy-budgeted	Yes	Yes (lattice-based)
WaterDrum [4]	LLM unlearning verification	Yes, single canonical query	Not addressed	Partially (non-adaptive server)	No
SEAL / SEAL-W	Both settings above	Yes	Yes, tunable (ϵ, δ)	Yes	Yes (Paillier, SEAL-W)

PIR [7] for RESPIR’s lattice-based construction. This is a deliberate simplification, not an oversight: lattice-based PIR is designed for sublinear communication at large database sizes, which matters for RESPIR’s fairness-audit setting (potentially large candidate pools of real users) but not at the batch sizes used here (tens of candidates per audit round), where Paillier’s linear server-side work and one-ciphertext-per-index communication are entirely practical and rest on a simpler, more standard hardness assumption (Section 4.1, Theorem 1).

2.4 Differential Privacy Accounting and Auditing

Channel B’s privacy cost is accounted via zero-concentrated differential privacy (zCDP) [6], chosen for its clean, tight composition over the B independent probe queries within one audit round. The Gaussian differential privacy / f -DP framework of Dong, Roth, and Su [12] gives the general Neyman–Pearson-optimal trade-off curve between a hypothesis test’s power and its leakage for a mechanism whose noise distribution is known exactly to the auditor – of which SEAL’s actual setting is a strictly harder special case, since the auditor knows only a finite calibration sample standing in for that distribution, not the distribution itself. We do not claim to solve, or even fully characterize, this harder finite-calibration variant of the f -DP tradeoff in general; Section 6.2 states this gap as an open question rather than papering over it with an unproven claim of optimality.

2.5 Influence-Based Spoofing Detection

Channel B’s ability to catch narrow, targeted fine-tuning (rather than genuine retraining) that a zero-cost own-data check cannot rests on the same intuition as Koh and Liang’s influence functions [13]: honest retraining moves the loss on a forgotten record’s *neighborhood* in feature space, not only the exact record itself, because the record’s removal changes the model’s fit in a region, not at a point; a narrowly targeted fine-tune that memorizes only the exact forgotten records does not reproduce this neighborhood effect. SEAL’s boundary-probe construction (perturbing D_f into nearby synthetic points, Section 3.2) operationalizes this intuition as a statistical test rather than as an influence-function computation, which would itself require access to the model’s internals the data owner does not have.

TABLE 2
Notation used throughout the paper.

D_f, D_r	Forgotten / retained record sets, $D = D_f \cup D_r$
θ, θ'	Model parameters before / after the claimed unlearning step
$l(\theta, x, y)$	Per-sample loss
Λ_A, Λ_B	SEAL Channel A / Channel B audit statistics
(ϵ, δ)	Differential-privacy budget spent by Channel B
$\sigma, L_{\max}, B$	Channel B noise scale, loss-clip bound, probe count
R (classical)	Number of calibration replicates (Section 3.2)
R (SEAL-W)	Real-cluster / phantom-cluster size (Section 3.3)
m, N, P	Tampered / total / checked slots (Theorem 2)
$\beta_{\text{target}}, \beta_{\text{ch}}$	Target and per-channel (Bonferroni-split) false-accusation rate
$\tau, q(\beta, m)$	Calibrated threshold, exact predictive quantile (Theorem 3)
z, z_i^*	Watermark z -score; the one PIR-retrieved real-cluster score

3 SYSTEM DESIGN

3.1 Threat Model and Problem Formulation

3.1.1 Setting and Notation

A data owner (or a regulator acting on its behalf) submitted a request to erase a record set D_f from a model trained by an operator (“the server”) on $D = D_f \cup D_r$, where D_r is the retained set. The server returns a model M' (parameters θ' ; the model before the request had parameters θ) and claims D_f ’s influence has been removed. The data owner does not trust this claim, cannot inspect M' ’s weights, training log, or optimizer state, and interacts with M' only through whatever query interface the deployment already exposes (label/logit queries for the classical track; text generation for the LLM track). The data owner possesses D_f in full (it is, definitionally, the data whose erasure it requested) but does *not* possess, and is not entitled to learn anything further about, D_r . Table 2 collects the symbols used throughout the rest of the paper.

3.1.2 Adversary Classes

Table 3 lists the server behaviors in scope, in increasing order of sophistication. Each is a strategy for producing M' (and, for SEAL-W, an accompanying response policy) given full knowledge of the audit protocol’s construction –

TABLE 3
Adversary classes in scope, and which channel targets each.

Behavior	Description	Targeted by
Honest	Genuinely retrains/fine-tunes excluding D_f	(the null hypothesis every channel is calibrated against)
Lazy	Ignores the request; keeps training on all of D , D_f included	Channel A / SEAL-W slot channel
Targeted spoof	Fine-tunes only on D_f (classical track) to fool an anticipated own-data check, leaving D_r -wide influence untouched	Channel B (boundary probes detect the missing neighborhood effect)
Filter-only (LLM)	Output-filters the one exact known canonical audit string	SEAL-W real-cluster (any paraphrase defeats it)
Topic-filter (LLM)	Output-filters <i>any</i> phrasing mentioning the forgotten entity, uniformly	SEAL-W diversity channel (Prop. 1)
Partial topic-filter (LLM)	Output-filters a guessed subset of m known paraphrasings, resource-bounded	SEAL-W slot channel (Thm. 2)

a Kerckhoffs’s-principle-style assumption: only the calibration data and the auditor’s specific random draws in a given round are secret, not the protocol’s design.

Two adversary classes are explicitly *out* of scope, stated here rather than left implicit: a server that adapts its training or filtering strategy *after* observing which specific audit round it is being tested in (SEAL-W’s PIR obliviousness, Theorem 1, is precisely what rules this out within a round, but a server that learns something from the aggregate pattern of many audit rounds over time is not analyzed here); and a server with a costless, perfect semantic classifier willing to refuse *all* queries – audit and genuine alike – about an entity, discussed further in Section 6.1.

3.1.3 Design Goal

The data owner’s goal is a decision rule T , built only from D_f , a disjoint calibration corpus, and query access to M' , that:

- 1) is calibrated to a stated false-accusation rate β : $\Pr[T = \text{dishonest} \mid \text{honest}] \leq \beta$;
- 2) carries a stated, and where possible zero, bound on how much running T reveals about D_r – the quantity Tang et al.’s theorem [2] shows cannot be driven to zero *in general* for a sufficiently powerful behavioral test, but which SEAL and SEAL-W make an explicit, chosen dial rather than an unacknowledged side effect.

3.2 SEAL: Auditing Classical Federated Unlearning

3.2.1 Two-Channel Certificate

Let θ, θ' be the model’s parameters before and after the claimed unlearning step, and $l(\theta, x, y)$ the per-sample loss.

Channel A (zero cost). The data owner computes $\Lambda_A = \text{mean}_{(x,y) \in D_f} [l(\theta', x, y) - l(\theta, x, y)]$ entirely on data it already possesses. No noise is added and no privacy budget is spent, because the released quantity is a function only of the requester’s own data and the public model.

Channel B (tunable cost). The data owner perturbs its own forgotten records into synthetic boundary probes $x_i + \epsilon_j$, $\epsilon_j \sim \mathcal{N}(0, \text{jitter}^2 I)$, and asks the server to evaluate M' on them, label held fixed. Each returned per-probe loss is clipped to $[0, L_{\max}]$ and released through a Gaussian mechanism at noise level σ , giving Λ_B analogous to Λ_A but over the noised probe losses. Because the probes are synthetic points constructed by the auditor from its own data – not real retained records – the Gaussian mechanism’s sensitivity bound applies directly, and zCDP composition over B probes gives $\rho_{\text{total}} = BL_{\max}^2/(2\sigma^2)$, converted to (ϵ, δ) via $\epsilon = \rho_{\text{total}} + 2\sqrt{\rho_{\text{total}} \ln(1/\delta)}$ [6]. Given a target (ϵ, δ) , this inverts to the noise level σ the server’s responses must carry – turning the impossibility of [2] into a dial: choosing ϵ fixes exactly how much retained-set privacy this audit round may cost.

3.2.2 Calibration

Both channels are compared against a null distribution built by literally re-executing the honest procedure R times, on a disjoint public calibration corpus standing in for D_f , using the *same base random seed and per-client ordering* as the real training run. This detail is not cosmetic: warm-starting a calibration replicate from an already-converged model, or varying the base seed across replicates, under-integrates or adds spurious trajectory variance to the null and produces a degenerate, near-zero-variance threshold in our own early implementation attempts (Section 4.1 gives the corrected, exact threshold this null feeds into).

3.2.3 Certificate Algorithm

Algorithm 1 summarizes the full data-owner-side procedure. The server-side interface is minimal and identical across every honest or dishonest mechanism in Table 3: given a batch of inputs, return M' ’s per-sample losses. This is deliberate – the certificate’s power comes entirely from which inputs are queried and how the responses are combined, not from any cooperation the server must specially provide, so an honest server’s implementation burden is exactly the burden of answering ordinary inference queries.

3.3 SEAL-W: Auditing LLM Unlearning

A data owner contributed watermarked text (via [8], keyed per contributor) that was used to fine-tune a language model, and wants to check a claimed erasure of its influence.

3.3.1 Why a Cluster, Not One Query

A server recognizing one exact known trigger string is defeated by any paraphrase. A smarter server can instead recognize the *topic* (the forgotten entity’s name, however worded) and either suppress every phrasing uniformly, or hardcode canned replies for only a guessed subset of phrasings it can afford. Neither a single paraphrased query nor PIR alone defeats either variant: against uniform suppression there is no “which slot is checked” secret worth hiding, since the server treats every phrasing alike; against the resource-bounded guesser, hiding which slot is checked is exactly the right game, but requires more than one real query to hide among.

Algorithm 1 SEAL certificate (data-owner side)

Require: D_f , model access to θ (before) and θ' (after), calibration corpus D_c , target β_{target} , DP budget (ε, δ) , jitter scale, B probes, R calibration replicates

- 1: $\Lambda_A \leftarrow \text{mean}_{(x,y) \in D_f} [l(\theta', x, y) - l(\theta, x, y)]$ {Channel A, zero cost}
- 2: $\sigma \leftarrow \text{CALIBRATE\SIGMA}(\varepsilon, \delta, B, L_{\text{max}})$ {zCDP inversion, [6]}
- 3: Draw probes $\{x_i + \epsilon_j\}_{j=1}^B$, $\epsilon_j \sim \mathcal{N}(0, \text{jitter}^2 I)$
- 4: Query server for $\{l(\theta', x_i + \epsilon_j, y_i)\}$, clip to $[0, L_{\text{max}}]$
- 5: $\Lambda_B \leftarrow \text{mean}_j [\text{noisy_}l(\theta', \text{probe}_j) - l(\theta, \text{probe}_j)]$ {Channel B}
- 6: **for** $r = 1$ to R **do**
- 7: Retrain fresh from θ' 's base seed on D_c -substituted-for- D_f shard, same client ordering
- 8: Honestly fine-tune to exclude the substituted shard
- 9: Record $(\Lambda_A^{(r)}, \Lambda_B^{(r)})$ on this replicate
- 10: **end for**
- 11: $(\mu_A, \sigma_A), (\mu_B, \sigma_B) \leftarrow \text{mean/std of the } R \text{ replicate values}$
- 12: $\beta_{\text{ch}} \leftarrow \beta_{\text{target}}/2$ {Bonferroni split, Thm. 4}
- 13: $\tau_A \leftarrow \mu_A - q(\beta_{\text{ch}}, R) \sigma_A$; $\tau_B \leftarrow \mu_B - q(\beta_{\text{ch}}, R) \sigma_B$ {Thm. 3}
- 14: **return** **dishonest** if $\Lambda_A < \tau_A$ or $\Lambda_B < \tau_B$, else **honest**, with (ε, δ) spent and β_{target} controlled

3.3.2 Protocol

Commit. The auditor assembles a candidate batch: an R -member *real cluster* of independently-paraphrased queries about the forgotten entity, one positive-control canary query about a known-retained entity, an R -member *negative-canary cluster* of paraphrased queries about a phantom entity never present in any training data, and decoy queries about other retained entities. The server answers every query – it cannot yet tell which will be checked. Each answer's watermark z -score [8] and a 32-bit SimHash response fingerprint [14] are quantized to integers and committed via a SHA-256 hash over both arrays, so the server cannot revise an answer once it learns what matters.

Retrieve. The auditor retrieves, via real single-server Paillier PIR [7]: one uniformly-random real-cluster member's z -score (chosen only now, not at commit time), the positive canary, every negative-canary-cluster member (for the diversity baseline), and every real-cluster fingerprint.

Decide. Two independent channels, plus canary sanity checks. The *slot channel* flags if the one retrieved real-cluster z -score exceeds a calibrated upper threshold. The *diversity channel* flags if the real cluster's mean pairwise Hamming diversity, relative to the phantom cluster's, falls below a calibrated lower threshold. Both thresholds, and the two canary sanity bounds, are calibrated from held-out calibration authors exactly as in Section 3.2, with the false-accusation budget β_{target} split across all active conditions (Theorem 4). Algorithm 2 summarizes the full auditor-side procedure.

In the experiments of Section 5, $R = 5$ (the full paraphrase-template pool, so the hypergeometric prediction of Theorem 2 matches the resource-bounded adversary's guessed subset one-for-one), $n_{\text{decoys}} = 2$, and $R_{\text{calib}} = 2$ calibration authors – the minimum for a defined sample

Algorithm 2 SEAL-W certificate (auditor side)

Require: forgotten entity f , retained entities, phantom entity p , real-cluster size R , n_{decoys} , PIR keypair, calibration authors, β_{target}

- 1: **Commit** (server side): build batch = R paraphrases of $f \cup 1$ positive canary $\cup R$ paraphrases of $p \cup n_{\text{decoys}}$ decoys; shuffle
- 2: Server answers every prompt in the batch (cannot yet tell which will be checked)
- 3: For each answer: compute watermark z -score [8] and 32-bit SimHash fingerprint [14]; quantize both to integers
- 4: Commit = SHA-256(database || fingerprints)
- 5: **Retrieve** (auditor side, real Paillier PIR [7]): $i^* \sim \text{Uniform}(\{1, \dots, R\})$ real-cluster index; retrieve z_{i^*} , positive-canary z , all R negative-canary z 's and fingerprints, all R real-cluster fingerprints
- 6: **Decide:** build nulls $\{z^{(r)}, \text{div-ratio}^{(r)}, z_{\text{pos}}^{(r)}, z_{\text{neg}}^{(r)}\}_{r=1}^{R_{\text{calib}}}$ by re-running honest unlearning targeting each calibration author instead of f
- 7: $n \leftarrow$ number of conditions calibrated in this call (up to 4); $\beta_{\text{ch}} \leftarrow \beta_{\text{target}}/n$ (Thm. 4)
- 8: $\tau_{\text{slot}}, \tau_{\text{div}}, \tau_{\text{pos}}, \tau_{\text{neg}} \leftarrow$ predictive quantiles at β_{ch} from each null (Thm. 3)
- 9: **return** **dishonest** if $z_{i^*} > \tau_{\text{slot}}$ or $\text{div-ratio} < \tau_{\text{div}}$ or $z_{\text{pos}} < \tau_{\text{pos}}$ or $z_{\text{neg}} > \tau_{\text{neg}}$, else **honest**

variance, and, as Section 6.1 discusses at length, the binding constraint on this track's measured reliability.

3.3.3 Grounding the Author Pool in a Real Benchmark

The audited fictitious-author profiles can be built two ways. The default experiment invents author names and generates each author's watermarked training documents from scratch. A second, additive source instead grounds the same protocol in the real, published TOFU benchmark [5]: TOFU's "full" split lays out 200 fictitious authors as 4,000 question/answer rows in 200 contiguous 20-row blocks with no explicit author-id column, so real names and biographical facts must first be recovered from the raw text. We do this with two deterministic heuristics, hand-verified against three independently-known authors and swept across all 200 blocks: an author's name is the single most frequent two-word run of consecutive capitalized words across the block's text (TOFU answers restate the full name far more often than any incidental two-capitalized-word phrase, such as a book title), with a trailing possessive normalized so that "X" and "X's" mentions reinforce one tally rather than splitting it; a short fact snippet is the first sentence of the first answer matching a fixed keyword-priority order (genre, award, profession, notable work, birth date), skipping bare name restatements.

Critically, this does *not* mean training on TOFU's literal question/answer text: that text is static, pre-existing, and carries no watermark bias, so using it directly as a training document would produce a document with no green-list signal at all and silently defeat the entire certificate. Instead, the recovered name and fact snippet only seed the same, unmodified watermarked-generation call ([8], Section 3.3) that the default experiment uses with an invented name

– every training document is still our own watermarked generation, now about a real entity and a real fact rather than a fabricated one. The reserved negative-canary pool is drawn from the same real TOFU distribution and held out of all fine-tuning entirely across a run, rather than hand-invented, for the same reason: a genuine, distributionally-matched “never trained on” control is a strictly better negative canary than an author name chosen by the auditor with no textual grounding at all. This substitution changes only where the author pool’s content comes from, not the certificate, the mechanisms, or any threshold in Sections 3.2–4.1; it is not comparable to the official TOFU leaderboard, which fine-tunes and evaluates phi-1.5 or LLaMA-2-7B against TOFU’s own forget-quality metrics, rather than *distilgpt2* against a watermark/PIR certificate.

4 THEORETICAL GUARANTEES AND COMPARISON TO PRIOR ART

4.1 Theoretical Guarantees

We state five results precisely; complete proofs are in the supplemental material. Each states exactly, and only, what it establishes.

Theorem 1 (PIR obliviousness). *If Paillier is IND-CPA secure (equivalently, the Decisional Composite Residuosity assumption holds), then for any two indices $i \neq j$ in an n -slot database, the one-hot-encrypted PIR query for i is computationally indistinguishable from that for j , with $\text{Adv}_{\text{index-prio}}(A) \leq 2 \text{Adv}_{\text{IND-CPA}}(B)$ for an adversary B built from A via a two-hybrid argument.*

Theorem 2 (Hypergeometric slot-hiding bound). *For a batch of N slots, P of which the auditor checks (uniformly at random, independent of the adversary’s choice), and an adversary that tampers with m slots before learning which will be checked,*

$$\Pr[\text{checked slots} \cap \text{tampered slots} = \emptyset] = \binom{N-m}{P} / \binom{N}{P}.$$

Theorem 3 (Exact finite-sample calibrated threshold). *Let $X_1, \dots, X_m \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d. (calibration replicates) and $Y \sim \mathcal{N}(\mu, \sigma^2)$ independent (the real audit statistic under the null). Then $(Y - \bar{X}) / (S\sqrt{1+1/m}) \sim t_{m-1}$, so $\tau = \bar{X} + t_{m-1, 1-\beta} S\sqrt{1+1/m}$ satisfies $\Pr[Y > \tau] = \beta$ exactly. This threshold strictly exceeds the asymptotic z -quantile plug-in at every finite m , and converges to it as $m \rightarrow \infty$.*

Proposition 1 (Exact zero diversity under uniform suppression). *If a server answers every query mentioning a topic with one fixed string, then for any real-cluster retrieval of $R \geq 2$ such queries, the mean pairwise SimHash Hamming diversity is exactly 0.*

Theorem 4 (Bonferroni composition, and the bug it caught). *If a certificate’s dishonesty decision is the logical OR of k independently-calibrated conditions each controlled at level β_i , the overall false-accusation rate is bounded by $\sum_i \beta_i$ (Boole’s inequality), so a target rate β_{target} requires $\beta_i = \beta_{\text{target}}/k$. Our first SEAL-W implementation calibrated all $k=4$ conditions (slot, diversity, two canary checks) at the same, un-split β_{target} , bounding the true rate only at $4\beta_{\text{target}}$ – a gap found by writing this theorem down and checking the code against it, not by an*

TABLE 4
WaterDrum-equivalent baseline vs. full SEAL-W against *topic_filter*, per trial ($N_{\text{calib}} = 2, \beta_{\text{target}} = 0.10$).

Trial	Real-cluster div.	SEAL-W	WaterDrum-baseline
0	0.0 (thr. -1.02)	No	Yes (also flags honest)
1	0.0 (thr. -0.38)	No	No
2	0.0 (thr. $+0.46$)	Yes	No

experiment failing visibly, and fixed prior to the experiments reported here.

4.2 Reproducing and Breaking WaterDrum

WaterDrum’s verification methodology – watermark a contribution, then test whether post-unlearning generations still carry the watermark – is, architecturally, exactly SEAL-W’s slot channel run in isolation: one canonical (unparaphrased) query, one watermark score, one calibrated threshold, no diversity channel, no real-cluster. We implement this literal reduction as an explicit baseline (*waterdrum_baseline_decide*) and run it side by side with the full SEAL-W decision on the same fine-tuned model and the same four server behaviors, honestly noting one substitution: WaterDrum’s own implementation embeds its watermark via Waterfall’s paraphraser-and-permutation scheme [11], while we use the simpler Kirchenbauer green-list watermark [8] throughout, for tractability on the CPU-only hardware used in our experiments. This substitution does not change the point demonstrated below: the failure mode is about a server detecting and filtering the audit *query*, not about either watermark’s own robustness to paraphrasing.

Table 4 reports the result against a server that uniformly suppresses every query mentioning the forgotten entity’s name (*topic_filter*). Every trial shows the real cluster’s response diversity collapsing to *exactly* zero (Proposition 1 proves this must happen, not merely that it tends to), and in the one trial where the calibration sample was not itself too degenerate to support any confident call, SEAL-W’s diversity channel flags the attack while the WaterDrum-equivalent single-query check does not – reading a filtered, non-watermarked canned reply as indistinguishable from genuine forgetting. This is precisely the false negative WaterDrum’s Appendix D reports as unsolved, reproduced directly against the mechanism built to exploit it, not merely cited.

Trials 0 and 1 illustrate a second, independent finding: at $N_{\text{calib}} = 2$, the Bonferroni-corrected threshold (Theorem 4) can itself land below zero, so even the true, exact-zero diversity ratio this adversary always produces is not always low enough to cross it. This is not a defect specific to WaterDrum-style checks; it is the same finite-calibration-sample limitation that governs every threshold in this paper, surfacing here in the one comparison built specifically to expose it.

5 EXPERIMENTAL EVALUATION

5.1 Classical Track

Federated multinomial-logistic/softmax models are trained across 5 simulated clients on three datasets of increasing

TABLE 5

Detection power vs. lazy noncompliance (Channel A alone, zero privacy cost) vs. two baselines.

Dataset	Param.-diff	Unrestr. behav.	SEAL (Ch. A)
German Credit	0.743	0.553	0.920
MNIST	0.597	0.500 (chance)	1.000
Covertime (150k)	0.610	0.510	0.540

scale: UCI German Credit (a bank-consortium right-to-be-forgotten instantiation, ~ 40 -record forget sets), MNIST (5,000-sample subset, 784-dim), and a 150,000-row subsample of UCI Forest Covertime (54-dim, $\sim 30,000$ rows/client). Table 5 reports detection power against *lazy* noncompliance (power at which SEAL’s zero-cost Channel A alone separates honest from lazy) against two baselines: raw parameter distance, and an unrestricted behavioral audit in the style of [2].

At higher input dimension (MNIST), the unrestricted behavioral baseline performs at exact chance: broad queries over a 784-dimensional retained population dilute one client’s ~ 50 forgotten records into noise, exactly the diffuseness [2]’s theorem predicts costs power. SEAL’s zero-cost channel needs no such broad query and reaches 1.000.

Table 6 reports the designed power/privacy-leakage tradeoff against *targeted spoofing* (fine-tuning only on D_f), across a sweep of Channel B’s privacy budget ε , together with retained-set leakage measured as the empirical membership-inference attack AUC an adversary achieves given exactly what Channel B releases. Concretely, the “attacker” here is not a separately trained classifier but the released statistic itself, read as a score: the (possibly noised) loss-gap $l(\theta', x, y) - l(\theta, x, y)$ is computed on a held-out pool of same-distribution, never-trained-on records (the “non-members”) and on genuinely retained records (the “members”), and the AUC of separating the two using exactly that one released number *is* the leakage measure – an honestly worst-case-for-the-defender proxy, since it hands the adversary the released quantity directly rather than requiring it to be learned from limited access. SEAL’s power climbs monotonically with ε toward the unbounded-noise limit, while leakage stays close to chance (0.50–0.54) across the whole sweep on German Credit and MNIST – with the label-preserving synthetic probes used here, the near-neighbor membership signal is intrinsically weak at these datasets’ scale (small forget sets, ~ 40 –50 records), so a tight ε does not have to suppress much signal in the first place. Parameter-diff also catches this specific spoof at zero accounted privacy cost, but only because the spoof mechanism used here is an aggressive, large, narrowly-directed parameter update; a subtler spoof would not show up in parameter distance at all, and there is no way to know which regime a real deployment is in without an audit like Channel B’s.

Figure 1 plots detection power against retained-set leakage directly for all three datasets and both baselines, making the same comparison visually: SEAL’s own-data channel (against lazy noncompliance) sits at zero measured leakage cost by construction, strictly above both baselines at every leakage level they occupy, on German Credit and MNIST;

TABLE 6

SEAL Channel B: power vs. targeted spoofing and retained-set leakage (attack AUC), across privacy budget ε .

ε	German Credit	MNIST	Leakage (both)
0.05	0.525	0.542	0.50–0.53
0.50	0.528	0.556	0.50–0.53
1.0	0.538	0.556	0.50–0.53
4.0	0.583	0.653	0.50–0.53
10^6 (no noise)	1.000	1.000	0.50–0.53

TABLE 7

Successive certificate fixes, same experiment (3 trials, $N_{\text{calib}}=2$).

Version	False-acc. rate	Detect. topic-filt.
Hardcoded canary floor	1.0 (degenerate)	1.0
Calibrated, asymptotic z	0.667	1.0
Calibrated, exact Student- t	0.333	0.667
+ Bonferroni split (Thm. 4)	0.0	0.333

against targeted spoofing, SEAL’s curve sweeps from near-chance power at tight ε to the two baselines’ fixed operating point as ε relaxes, tracing out the tradeoff Table 6 reports numerically.

Covertime’s result is reported as a genuine, honest limitation, not tuned away: at $\sim 30,000$ rows/client, SEAL’s own-data channel loses most of its power (0.540, barely above the 0.500 chance line and the two baselines), because a federation with far more data per client is intrinsically more robust to losing any one client’s contribution – the audit signal “did removing this data change how well the model fits it” is weakest exactly where the model has the most other data to fall back on. This is visible directly in Figure 1’s bottom panel: unlike the top two datasets, SEAL’s lazy-detection point is barely separated from the baselines rather than sitting in a clearly dominant region.

The certificate’s calibrated single-instance decision (Theorem 3’s exact threshold) is more conservative than the raw statistic: German Credit’s false-accusation rate fell from 0.15 to 0.10 and MNIST’s from 0.083 to 0.0 after replacing the asymptotic z -quantile plug-in with the exact predictive quantile, with no change to the Monte-Carlo power numbers above (those do not depend on the threshold formula).

5.2 LLM Track

distilgpt2 is fine-tuned (CPU-only) on watermarked biographies of fictitious authors drawn from the real TOFU benchmark [5], audited by SEAL-W against four server behaviors of increasing sophistication (*lazy*, exact-string *filter-only*, uniform *topic-filter*, and a resource-bounded *partial topic-filter* that guesses 50% of a 5-member paraphrase pool). Table 7 tracks four successive, honestly-measured threshold fixes on the same experiment.

Each fix moves both numbers in the direction the corresponding theorem predicts *before* it was run: an exact threshold cannot manufacture confidence a small calibration sample does not support, so tightening the threshold to be exact, and then to be correctly composed across channels, necessarily costs detection power as it removes false accusations.

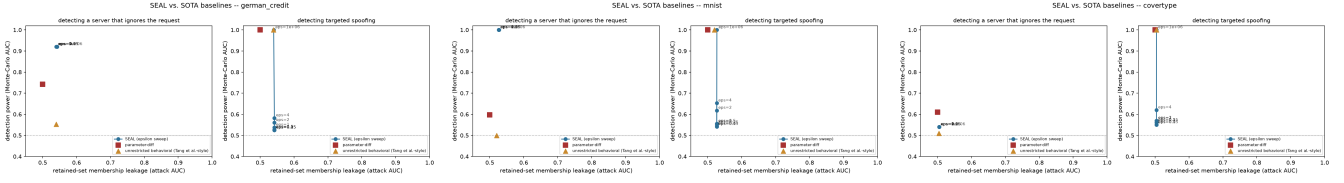


Fig. 1. Detection power vs. retained-set leakage (attack AUC), SEAL’s ε -sweep against both baselines: German Credit (left), MNIST (center), Covertypes (right). Each panel’s left subplot is against lazy noncompliance, right subplot against targeted spoofing.

TABLE 8

Full SEAL-W certificate, final (Bonferroni-corrected) version, all four server behaviors (3 trials, $N_{\text{calib}}=2$, $\beta_{\text{target}}=0.10$).

Server behavior	Detection rate
Honest (false-accusation rate)	0.0
Lazy (keeps training on everything)	0.333
Filter-only (exact-string canned reply)	0.667
Topic-filter (uniform suppression)	0.333
– diversity channel alone	0.333
Partial topic-filter (guesses 50% of $R=5$)	0.333
– analytic evasion probability	0.6

Table 8 reports the final, Bonferroni-corrected certificate’s detection rate against all four server behaviors, together with the diversity channel’s rate in isolation and the resource-bounded adversary’s analytic evasion prediction. The partial-topic-filter adversary guesses $m = 2$ of the $R = 5$ paraphrase templates (a guess fraction of 0.5, rounded half-to-even by Python’s `round`, so $0.5 \times 5 = 2.5$ rounds to the even integer 2, not 3); the resulting empirical detection rate (0.333) is consistent with the exact analytic hypergeometric prediction (Theorem 2: $1 - (R - m)/R = 1 - 3/5 = 0.4$) within this trial count’s sampling noise – the one place in this evaluation where a component is checked against a closed-form formula rather than only Monte Carlo. The WaterDrum-comparison result (Section 4.2, Table 4) is drawn from this same final run.

This 3-trial, $N_{\text{calib}}=2$ regime is small by design: each trial fine-tunes and audits a real language model on CPU-only hardware, and every rate above should be read as “roughly this many out of 3,” not a precise estimate – the qualitative pattern (which mechanism is caught by which channel, and that both fixes moved every rate in the theoretically-predicted direction) is the result this scale supports, not the exact percentages. Section 6.1 returns to this as the paper’s central open lever.

5.3 Implementation and Reproducibility

Every mechanism, channel, and adversary discussed above is implemented, not simulated: the federated softmax model and both classical-track channels are pure-Python/NumPy; the LLM track fine-tunes a real `distilgpt2` checkpoint via PyTorch/Hugging Face Transformers; the watermark is the published green-list construction [8], not a proxy for it; and SEAL-W’s PIR is a real Paillier implementation (the `phe` library), with the auditor’s queries genuinely encrypted end-to-end rather than the obliviousness property being asserted without a corresponding cryptographic artifact. All four adversary mechanisms (lazy, filter-

TABLE 9

Classical-track (SEAL) hyperparameters, exact per dataset, sufficient to reproduce every number in Tables 5–6.

Parameter	German Credit	MNIST	Covertypes
Clients	5	5	5
FedAvg rounds (train)	25	20	20
Unlearn extra rounds	8	6	6
Local epochs, lr	1, 0.3	1, 0.3	1, 0.3
Batch size	32	64	512
Calibration replicates R	25	15	12
Probe count B , jitter	50, 0.4	50, 0.05	50, 0.3
Monte-Carlo trials	20	12	10
β_{target}	0.05	0.05	0.05

TABLE 10

LLM-track (SEAL-W) hyperparameters, sufficient to reproduce every number in Tables 7–8.

Base model	<code>distilgpt2</code> (82M params, CPU-only)
Fine-tune epochs, lr, doc repeats	$3, 5 \times 10^{-5}, 8$
Watermark γ (green fraction), δ (logit bias)	0.5, 8.0
Documents/author, generation length	4, 30 tokens
Authors/trial, calibration authors	5, 2
Real-cluster / phantom-cluster size R	5, 5
Decoys, Monte-Carlo trials	2, 3
β_{target}	0.10, split across up to 4 channels

only, topic-filter, partial-topic-filter) share identical underlying model weights within each experiment – a single “kept training on everything, never actually unlearned” fine-tune – differing only in the output-filtering wrapper placed over it, so that the comparison isolates what each certificate channel can detect rather than accidentally comparing different models. The full implementation, experiment scripts, and raw per-trial result files are available at <https://github.com/vinhqdgang/SEAL-Statistically-bounded-Erasure-Audit-under-mutual-distrust>; the paper’s numbered theorem statements are additionally restated with complete, checkable proofs in the supplemental material. Tables 9–10 list every hyperparameter needed to reproduce the numbers in this section exactly, including per-dataset values that differ across the classical-track sweep.

A brief note on dual use: publishing a concrete audit protocol necessarily also publishes what it does and does not catch (Section 3.1’s scoped-out adversary classes, Section 6.1’s explicit caveats). We consider this a feature, not a risk to withhold – an audit whose coverage is secret cannot be trusted by the mutually distrusting party it is meant

to serve, and every adversary class this paper evaluates is already a natural strategy a self-interested operator would consider without reading this paper first. The construction does not itself enable any new attack on a model or its training data; it is strictly a verification tool run by the data owner’s side of the protocol.

6 DISCUSSION, FUTURE WORK, AND CONCLUSION

6.1 Discussion and Limitations

6.1.1 Calibration Sample Size Is the Binding Constraint

The remaining gap between SEAL(-W)’s design and its measured single-instance reliability is governed by calibration-sample size, not by the threshold formula. Theorem 3’s correction is exact given the Gaussian idealization of the calibration replicates, but at $N_{\text{calib}} = 2$ (this paper’s LLM-track minimum, needed just to define a sample variance at all) which two replicates happen to be drawn dominates the resulting threshold more than the mechanism under test does – visible directly in Table 4’s per-trial spread, where the same true, exact-zero diversity signal is caught in one trial and missed in the other two purely because the calibrated threshold itself moved. Every fix applied in this paper (Table 7) made the certificate’s stated guarantee *more* truthful and, as a direct and expected consequence, less powerful at this sample size; none of them, and no other threshold formula, can substitute for more replicates. This is not a defect specific to the LLM track: the classical track’s Channel B shows the same pattern at its own, larger but still finite R (Section 3.2), and Coverttype’s result below shows a second, independent way finite information can bind.

6.1.2 Scale Is Not a Free Win

Coverttype’s power loss is a genuine, dataset-scale-dependent limitation of the own-data channel, not a universal one, and is reported as such rather than tuned away by choosing a friendlier configuration. The mechanism is intuitive in retrospect but was not assumed going in: a federation with $\sim 200\times$ more rows per client than German Credit converges to a substantially more stable optimum, so continuing to train on everything (the “lazy” adversary) perturbs the model’s fit on any one client’s forgotten records by much less, relative to ordinary round-to-round training noise at that scale, than it does with a few hundred rows per client. The audit signal Channel A relies on – “did removing this data change how well the model fits it” – is intrinsically weakest exactly where the model has the most other data to fall back on. Whether a tunable-cost channel analogous to Channel B, or a larger calibration budget, restores power at this scale is left as a direct, concrete next experiment rather than a claim made here.

6.1.3 What No Channel in This Paper Rules Out

Neither SEAL-W channel defeats a hypothetical server with a perfect, costless semantic classifier willing to refuse *all* queries – ordinary and audit alike – about an entity, i.e. a server behaviorally indistinguishable from a total, standing topic ban rather than a targeted, audit-specific spoof. No

black-box behavioral test, with or without PIR-hidden retrieval, can formally rule this out: if the server is willing to make M' refuse to discuss an entity under any circumstance forever, no query-only audit can distinguish “never learned about this entity” from “knows about it and refuses to ever say so.” This is the same scope boundary Tang et al.’s impossibility framing draws for behavioral audits in general [2], and the same kind RESPIR’s own theorem statements draw for their PIR construction [9] – stating the covered adversary class precisely, rather than an unconditional claim, is the standard this literature already holds itself to.

A second, practical caveat concerns the diversity channel specifically: Proposition 1 is exact for literal, single-string uniform suppression, but a server that adds cheap, content-preserving noise to an otherwise-canned reply (paraphrasing its own filler text, injecting innocuous filler tokens) could in principle defeat a naive SimHash-based diversity check without restoring genuine informativeness, without being caught by the specific guarantee proved here. Studying this specific arms race – diversity-channel robustness against a server that knows it is being diversity-checked – is out of scope for this paper, which evaluates the concrete adversary it implements and measures (uniform canned-reply suppression), not every conceivable output-filtering strategy.

6.2 Future Work: An Open Theoretical Question

The certificate design in Sections 3.2–3.3 is engineering, in the sense used throughout this paper: a specific, checkable composition of known primitives, not a new theorem about the fundamental limits of calibration-based auditing. A genuinely open, and considerably harder, question is whether those limits admit a clean characterization at all. We state it here as a research agenda, not a result, and explicitly flag any future claim to have solved it in a single sitting as suspect until checked line by line.

The calibration-only erasure audit game. A server holds $D = D_f \cup D_r$ and an unlearning claim for D_f . The auditor does not know D_r (the entire point of mutual distrust) or the server’s mechanism family, and knows only D_f , a disjoint calibration corpus D_c of size m , and query access to M' . The auditor builds a test using only these, calibrating thresholds empirically from replicated honest runs on D_c – exactly what both certificates in this paper already do. Two quantities characterize the test jointly, as functions of the calibration budget m and query budget q : power $P(m, q)$ at a controlled false-accusation rate, and aggregate retained-set leakage $L(m, q)$, against a server that is not honest-but-curious but adversarially adaptive, choosing how to respond to every query to best evade the test while knowing the test’s construction (though not its specific random draws).

This formalization is deliberately posed to specialize back to the known frameworks under extra assumptions: if the server is honest-but-curious and D_r reduces to a single known point, it should recover Tang et al.’s bound [2]; if the mechanism’s noise family is known exactly and $m \rightarrow \infty$, it should recover the f -DP tradeoff function [12]. Verifying these reductions is itself the first, necessary step – if the formalization does not specialize correctly, it is the wrong model and needs revision before anything built on it is worth trusting.

Three concrete conjectures organize an attack on this question, in increasing order of difficulty and decreasing order of how well existing tools apply: (A) *distribution-free achievability* – a calibration-only test achieving power within $O(1/\sqrt{m})$ of an oracle test that knows the mechanism’s noise distribution exactly, via standard concentration inequalities on the empirical calibration mean/variance; (B) *a matching minimax lower bound* against the adaptive, protocol-aware adversary above, via a Le Cam- or Fano-style argument where the two hypotheses being distinguished are two *server strategies* rather than two datasets – the least well-trodden of the three, and the one most likely to require a genuinely new argument; and (C) *adversarial multi-query composition*, treating the audit as a repeated game between an auditor allocating a continuous falsification-response budget and a no-regret learner, closer to online-learning regret bounds than to classical hypothesis testing. The most likely outcome of attempting (A) is that it reduces to a known concentration-inequality corollary once correctly stated, which would still be a real, if modest, contribution (the formalization itself); the most interesting possible outcome of (B) is a clean impossibility result with an explicit, non-obvious dependence on m . Either way, this is realistically months of focused work by someone with a research background in hypothesis testing or differential-privacy theory, and is offered here as a scoped, falsifiable agenda for that work, not a preview of its answer.

6.3 Conclusion

SEAL and SEAL-W give a mutually distrusting data owner and model operator a concrete, calibrated, formally-scoped audit certificate where prior work characterized the problem’s difficulty without a mechanism to run. Reproducing WaterDrum’s verification methodology as an explicit baseline shows it inherits a specific, previously-only-stated blind spot that SEAL-W’s independent diversity channel closes; five proven theorems state precisely, and only, what is guaranteed; and the experimental evaluation reports where the certificate’s power degrades – at large per-client data scale, and at small calibration-sample size – as plainly as where it succeeds.

REFERENCES

- [1] European Parliament and Council of the European Union, “Regulation (eu) 2016/679 (general data protection regulation),” Official Journal of the European Union L119, 2016.
- [2] L. Tang, J. Joshi, and A. Kundu, “Characterizing privacy-audit alignment in behavioral audit of machine unlearning,” *arXiv preprint arXiv:2606.14518*, 2026, v2.
- [3] T. L. Nguyen, M. Tuler de Oliveira, A. Braeken, A. Y. Ding, and Q.-V. Pham, “Towards verifiable federated unlearning: Framework, challenges, and the road ahead,” *IEEE Internet Computing*, 2026, arXiv:2510.00833.
- [4] X. Lu, X. Niu, G. K. R. Lau, B. T. C. Nhung, R. H. L. Sim, J. R. Himawan, F. Wen, C.-S. Foo, S.-K. Ng, and B. K. H. Low, “Waterdrum: Watermarking for data-centric unlearning metric,” in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2026, arXiv:2505.05064.
- [5] P. Maini, Z. Feng, A. Schwarzschild, Z. C. Lipton, and J. Z. Kolter, “Tofu: A task of fictitious unlearning for llms,” in *Conference on Language Modeling (COLM)*, 2024.
- [6] M. Bun and T. Steinke, “Concentrated differential privacy: Simplifications, extensions, and lower bounds,” *Theory of Cryptography Conference (TCC)*, 2016.
- [7] P. Paillier, “Public-key cryptosystems based on composite degree residuosity classes,” in *Advances in Cryptology—EUROCRYPT ’99*. Springer, 1999, pp. 223–238.
- [8] J. Kirchenbauer, J. Geiping, Y. Wen, J. Katz, I. Miers, and T. Goldstein, “A watermark for large language models,” in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.
- [9] A. Godinot, S. Azogagh, J. Ferry, and S. Gambs, “Manipulation-proof oblivious audits against deceptive model providers,” *arXiv preprint arXiv:2608.04365*, 2026.
- [10] T. González, M. Dulce-Rubio, A. Ramdas, and M. Ribero, “Sequentially auditing differential privacy,” *arXiv preprint arXiv:2509.07055*, 2025.
- [11] G. K. R. Lau, X. Niu, H. Dao, J. Chen, C.-S. Foo, and B. K. H. Low, “Waterfall: Framework for robust and scalable text watermarking and provenance for llms,” in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024.
- [12] J. Dong, A. Roth, and W. J. Su, “Gaussian differential privacy,” *Journal of the Royal Statistical Society Series B*, vol. 84, no. 1, pp. 3–37, 2022.
- [13] P. W. Koh and P. Liang, “Understanding black-box predictions via influence functions,” in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017.
- [14] M. S. Charikar, “Similarity estimation techniques from rounding algorithms,” in *Proceedings of the 34th Annual ACM Symposium on Theory of Computing (STOC)*, 2002, pp. 380–388.